



Optimizing Renewable Energy Infrastructure via AI-Driven Predictive Maintenance With Fault Detection: Its Industrial Psychological Perspective

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Abstract: The unprecedented growth of renewable energy infrastructure poses both technical and humanistic issues, especially about the reliability of the system, efficiency, and adaptability of the workforce. This literature review investigates the role of artificial intelligence (AI) in optimizing renewable energy systems (RES) through fault detection and predictive maintenance (PdM) from an industrial psychology perspective to understand human-technology interactions. The research reflects current studies on AI-powered monitoring devices, anomaly detection methods, and PdM models, and their usefulness in minimizing unexpected downtime, extending equipment life, and optimizing overall equipment performance. At the same time, the review highlights how industrial-psychological influences, including employee attitudes, cognitive preparation, and behavioral adaptation, can determine the successful adoption and use of AI-enabled maintenance solutions. It has been shown that the benefits of AI technologies, development of trust, and reduction of resistance to technological change are highly dependent on workforce engagement, training, and participation in decision-making. The review also establishes the major organizational, social, and economic consequences of incorporating AI into human-centric responses, indicating that the best opportunities are expected to arise from matching technical advances with human behavioral concerns. Lastly, existing research gaps are addressed, including empirical research on the long-term relationship between humans and AI, cross-cultural workforce strategies, and scalable implementation plans for various renewable energy infrastructure. Altogether, this review has demonstrated the prospective changes that can be made to the possibility of stabilizing the renewable energy infrastructure through the combination of PdM, AI-based fault detection, and industry-oriented psychological insights, providing a holistic framework to improve the stability, sustainability, and human-machine interactions of the renewable energy infrastructure.

Keywords: Renewable energy; Artificial intelligence; Predictive maintenance; Fault detection; Industrial psychology

1 Introduction

The last few years have seen a boom in the renewable energy industry like never before, as people take climate change seriously and work toward a shared goal of reducing their dependence on fossil fuels [1]. However, the intermittency of renewable energy sources, including solar and wind, poses operational challenges and requires innovative solutions. The growing energy needs, combined with environmental issues, have initiated a worldwide transformation toward renewable energy sources. Solar, wind, and hydropower are more environmentally friendly options that will reduce reliance on nonrenewable fossil fuels and the environmental footprint of conventional energy sources [2]. Renewable energy sources have become more accessible and cheaper, making their adoption a common trend as renewable energy technologies continue to develop. Governmental and public research and development in the renewable energy domain will lead to improved efficiency and ensure that future energy demand is met due to low cost, ease of maintenance, durability, and infinite availability [3].

Renewable energy is beneficial not only for sustainability but also economically. It is advantageous to the economy because it helps reduce electricity production costs by using natural, renewable energy sources. It may

also be a by-product of income since users will be able to sell the electricity that they have generated to the power grid [4]. Most power generation still relies on fossil fuels, even though the use of renewable energy sources is on the rise, owing to the intermittency of renewable energy and the high upfront costs involved. For example, a photovoltaic system will only work during the day, a wind turbine will only work when there is sufficient airflow, and a hydro turbine will only work when there is sufficient water flow. The current paper discusses how artificial intelligence (AI) applies to the renewable energy industry and its potential for predictive maintenance (PdM) and energy optimization. According to Omerbegović and Omerbegović [5], the recent growth of renewable energy sources has spurred further research into addressing their shortcomings. Modern technologies, AI, and machine learning (ML) in particular, have significant potential to address the instability and uncertainty of renewable supply, which is a major concern, given that their basic goal is to process uncertain data. The author looks at how ML can be used in many realms of renewable energy systems (RES), including forecasting, where ML can be used to generate accurate models; maximum power point tracking, which can get a steady and smooth control that is not easily affected by input noise; and inverters, where ML can be applied to provide high-quality power with no fluctuation despite a choppy input. Although ML offers numerous possibilities for addressing challenges in RES, its performance in solving a specific system's challenges will largely depend on several factors. The author analyses all these topics by systematically exploring the merits and disadvantages of ML applications in hybrid RES.

The combination of ML and AI fully revolutionised the energy industry through the means of stabilising the grid, forecasting demands, and increasing the generation of renewable energy sources. Handling vast amounts of information related to energy requires data science, without which it is impossible to make accurate forecasts and provide information-based decision-making. This study examines the role of ML, AI, and data science in advancing renewable energy technology (RET). Among the research topics, it is possible to distinguish energy storage optimisation, smart grids, and PdM [6]. Key research findings in the field are presented in a thorough literature review, which also demonstrates how AI and ML are applied in predictive modelling and energy management. After a thorough examination of the outcomes of AI-driven models, the study methods section describes the data-driven strategies utilised to maximise energy use. The paper concludes with recommendations for future lines of inquiry, policy implications, and how AI-enhanced energy systems can contribute to a more sustainable and resilient energy future. AI and ML are crucial to the advancement of renewable energy, as they utilise data science to enhance energy consumption, distribution, and generation [7]. ML models improve solar and wind power efficiency using predictive analytics, which forecast energy output based on historical data, weather trends, and real-time inputs. By reducing energy waste, integrating various renewable sources, and balancing supply and demand, AI-driven algorithms enhance grid stability.

Although they have responded by accelerating the implementation, policy innovations such as feed-in tariffs and renewable portfolio standards necessitate local adjustments to the policy. Many economic barriers remain and would require innovative financing plans and investment rewards. The inclusion of the community and social acceptance are noted as important factors of success, promoting the need to involve everyone in the decision-making processes. The analysis of an introduction of the so-called the green energy as the source of ensuring the energy independence of Ukraine is the primary concern of strategic response to economic difficulties, and as well the possibilities of the renewable energy sources involvement into the national energy complex functioning and its influence on the energy security and sustainable development of Ukraine are also discussed by Shulzhenko et al. [8]. The authors involve modelling the economic influence of integrating renewable energy, assessing legislative initiatives through comparative legal analysis, and a systematic review of the potential of renewable energy sources. To assess the changes that will occur in the energy sector in the future, statistical analysis and forecasting methods also play a significant role in enhancing the country's energy balance by increasing the share of renewable energy sources and reducing dependence on imported energy sources. Rushchitskaya et al. [9] highlights the opportunities of solar, wind and bio-energy systems. They identify the potential of how the integration of renewable energy sources could enhance urban infrastructure sustainability. They examine the current use of renewable energy in urban environments, highlighting key parameters such as technological innovations and challenges. They consider best practices and innovative concepts to incorporate renewable energy in the urban systems. These results suggest that although renewable energy can play a significant role in the sustainability of cities, its implementation can only be achieved by addressing challenges related to legal frameworks, societal alignment, and infrastructural issues. Kataray et al. [10] state that Renewable energy, in addition to assisting in environmental protection, is the best solution to the greenhouse gas emission problem, which is manifesting itself through increasing amounts. The tendency of electrical grids is to use one-way systems where only a third of the energy in fuel is converted into electricity, and the waste heat is not recovered. The development and efficient management of renewable energy sources are two advantages of smart grids, which are regulated power networks. An in-depth analysis of the features of smart technologies, energy storage systems, demand-side management, communication technologies, grid security, and privacy is also provided. Important challenges with integrating renewable energy sources into smart grids are also highlighted. The communication network and demand-side management, combined with the right algorithms, are found to be crucial

for integrating the smart grid in the future.

Barman et al. [11] analyse the utility interest, smart charging strategies enabled by RE, four key components of electric vehicle (EV) charging infrastructure, and related potential and difficulties are all examined. First, the usage of renewable energy sources for EV charging is discussed, together with their universal adoption, benefits, and drawbacks, as well as the leading countries. Second, they offered a thorough examination of energy storage technologies, charging infrastructure, related power electronics, and smart grid connectivity to promote the use of RE in EVs. Third, they looked closely at the many smart charging techniques employed by the industry employing RE considering the most recent global trend in EV energy consumption. Finally, given the inherent difficulties in achieving a sustainable transition, they examine the technological prospects and challenges related to resource optimisation, grid integration, renovation, standardisation, maintenance, and network security. The research by Ahmed et al. [12] explored the huge shift of the centralised to the decentralised energy systems over the last few years. These systems are expected to address the local energy needs by tapping the locally available renewable resources. This strategy helps in accelerating the process of transition to zero carbon emissions, reducing complexity and costs, enhancing efficiency, local resilience, and energy independence. Community energy is a critical component and is supported by the involvement of citizens in the generation of electricity using renewable sources all over the world but more so in the European countries. Even the European Union with its social focus on innovation and citizen engagement recognizes the importance of energy communities in its latest energy plan. The concept of integrating local energy communities or local-based energy initiatives has attracted the attention of the entire world because of the benefits that the utilization of renewable energy sources brings to the economy, the environment, and efficiency.

In a bid to expound on the role of renewable energy as an ingredient to the actualization of Sustainable Development Goals (SDGs), Adanma and Ogunbiyi [13] review the impact of the application of renewable energy in the economy and the surroundings of different places across the world. A shift towards renewable energy sources is a way to address the drastic reduction in pollution and greenhouse effects, as well as play a major role in the market, which would also generate additional jobs and stabilize the energy prices. In addition to the role played by technological innovations in eradicating the current barriers to adoption, the analysis points to the importance of global treaties and governmental regulations in ensuring global transition to renewable energy. Majeed et al. [14] suggest using renewable energy as an alternative source of energy as a way of effectively controlling agricultural energy. It talks of the possible advantages, hurdles, and prospects of incorporating RET in farmlands. They can contribute by doing a detailed examination of the alternative source of energy and the way it can be properly harnessed in the management of energy. They give considerate information to the researchers and other stakeholders in the agricultural sector who wish to transform the industry into production of renewable energy sources, indicating both advantages and threats of any action. Agreeing on alternative types of energy in agricultural energy management can assist in reaching the decrease in greenhouse gas emissions, enhance energy efficiency, and food sustainability. There should be removal of policy, economic and technical barriers and incentives to share information and thereby increasing the ability of farmers and stakeholders to conduct efficiently should guarantee successful implementation. Bathaei and Štreimikienė [15] observe that sustainable agriculture needs energy to work; this energy may be supplied using Renewable energy and will reduce the damage to the environment posed by agriculture. It helps generation of energy by using agricultural waste and lowers the consumption of fossil fuels. The indicators on renewable energy should be ecological to ensure the sustainability of agriculture production systems. These measures include the aspects of economy, society, environment, institutions and technology. This research makes areas of priority informed by frequent updating indicators, and it allows sustainable agriculture as well as making strategic decisions.

In contrast to previous reviews, which mainly focus on the algorithms, applications and technical performance of AI-based PdM in RES, this systematic review takes a wider area into industrial psychological perspective by incorporating technological, human and organizational aspects. This study combines evidence from multiple sources to examine the influence of factors such as employee trust in AI, collaboration between humans and AI, technology acceptance, workforce competence, cognitive workload, safety culture, leadership support, organizational readiness, employee training and resistance to technological change on the implementation and effectiveness of AI for PdM and fault detection. Therefore, the novelty of this study is not only on a technology-based review but also, the study develops a framework for a human-centred socio-technical approach to optimizing the utilization of renewable energy infrastructure via AI-driving PdM with fault detection for sustainably operations. Therefore, the study has the following objectives are:

(i) Systematically carry out review of the use of AI-based PdM and fault detection solutions for optimizing renewable energy infrastructure with particular focus on their success in enhancing system reliability, operational efficiency, and maintenance decision making.

(ii) Explore industrial psychological issues that affect the adoption and implementation of AI systems for PdM in renewable energy companies, such as trust in AI, human-AI teamwork, employees' competence, organizations' readiness to change, and resistance to technological change.

(iii) Understand the research trends, research gaps, and future research directions in integrating AI technologies

and industrial psychology in sustainable renewable energy infrastructure management and organizational performance.

1.1 Problem Statement of the Literature Review Study

The shift to renewable energy around the world is gaining momentum as more people become aware of the risks of climate change, energy security and the need to attain SDGs. Wind, solar, hydropower and smart grid are some of the RES that are now integral parts of modern energy infrastructure [16]. But they are sensitive to unforeseen equipment malfunction, part deterioration and extreme operating conditions, leading to higher maintenance expenses, lower energy production and lower efficiency and system downtime. Therefore, the assurance of the reliability and optimum performance of the renewable energy infrastructure has emerged as a critical issue for research and practice. AI-based PdM and fault detection technologies have proven to be useful tools for enhancing the performance of RES. These technologies can utilize ML, deep learning, Internet of Things (IoT) devices, and complex data analysis, allowing for real-time monitoring of machine condition and early detection of faults, along with PdM planning. While many studies have already highlighted the technical advantages of AI over improving system reliability, lowering maintenance expenses, and increasing equipment lifespan, the current literature focuses more on technological solutions and disregards human and organizational aspects that impact the successful implementation of AI.

Many psychological dimensions of the industrial environment are crucial to the success of AI-powered maintenance systems, including trust in AI, human-AI interaction, AI technology acceptance, employee competence, organizational preparedness, managers’ support, and opposition to technology change. Yet, these factors are dispersed throughout engineering, management and psychological literature with no work to date that brings together both technological and behavioral aspects in one synthesis [17]. However, this knowledge gap impedes human-centric AI strategy development of renewable energy infrastructure management. Thus, the purpose of the present systematic literature review (SLR) is to draw out the existing scientific evidence on PdM and fault detection using AI from the perspective of an industrial psychologist, to reveal existing research gaps, and to suggest a unified framework for future research, industrial practice, and sustainable development of renewable energy infrastructures.

1.2 Conceptual Framework for the Study

The present invention is presented as an integrated conceptual framework for optimizing renewable energy infrastructure by implementing an integrated solution of AI-driven PdM and fault detection from an industrial psychological perspective as shown in Figure 1.

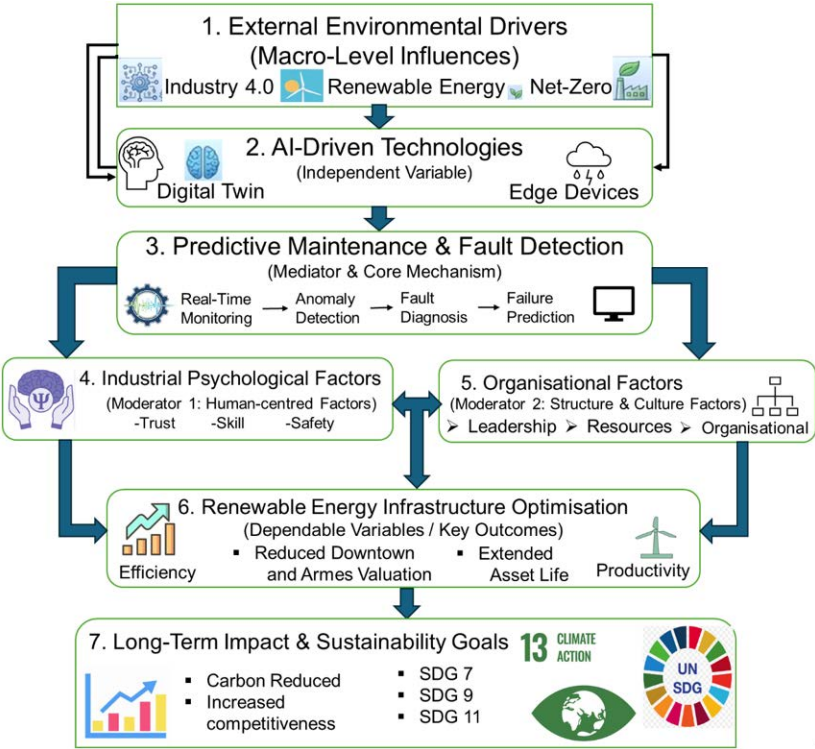


Figure 1. The conceptual framework for the optimization of renewable energy infrastructure for AI-driven predictive maintenance and fault detection, and its psychological perspectives

Note: UN = United Nations; SDGs = Sustainable Development Goals.

The framework starts by examining external environmental drivers—such as Industry 4.0, renewable energy transition and net-zero policies—that encourage the use of AI. PdM can be achieved using AI-driven technologies like digital twins and edge computing, which allow for real-time monitoring, anomaly detection, fault diagnosis, and failure prediction. These processes are enhanced by organizational factors (e.g., leadership, innovation, resources) and industrial psychological factors (e.g., trust, skills and safety) [15]. These interactions help to maximize infrastructure efficiency, minimize downtime and maintenance expenses, increase asset life, increase productivity and contribute to sustainability objectives, carbon reduction and the fulfilment of the United Nations SDGs.

2 Methodology

The methodology for this systematic review study cut across the research design, search strategy, inclusion, and exclusion criteria.

(i) **Research Design and Approach:** The present research has chosen a SLR that is developed according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework to achieve transparency, replicability, and rigor. To optimize the renewable energy infrastructure by utilizing the PdM and fault detection through AI: An Industrial Psychological Approach. PRISMA offers a systematic method that entails identification, screening, eligibility and inclusion of the studies.

(ii) **Search Strategy and Information Sources:** The search is performed in the key academic databases: Scopus, Web of Science, IEEE Xplore, ScienceDirect. Search strings are formulated in the form of Boolean operators: “AI-driven PdM” OR “ML maintenance” AND (renewable energy or smart grid or wind or solar) and (fault detection or anomaly detection) and (human factors or industrial psychology or human interaction). This ensures the combination of technical and psychological aspects.

(iii) **Inclusion and Exclusion Criteria:** Table 1 presents the study ways of analyzing articles from existing knowledge.

Table 1. Criteria for the inclusion of articles and exclusion of articles

Inclusion Criteria	Exclusion Criteria
Peer-reviewed journal articles from reputable outlet.	Non-English publications
Published between 2010–2026	Conference abstracts without full papers
Focus on AI, predictive maintenance (PdM), or fault detection	Studies unrelated to energy infrastructure
Studies addressing renewable energy systems (RES)	Purely theoretical AI papers without application context
Papers including human, organizational, or behavioral aspects	Papers of organization that does not involve in the AI, PdM, or fault detection

(iv) The PRISMA and Quality Assessment Selection of the Study Process

This research has the following four main steps in the PRISMA process shown in Figure 2: identification, screening, eligibility, and inclusion. Identification involves retrieving articles through database searches from Scopus, Web of Science, and Google Scholar. Screening involves removing duplicates and screening titles and abstracts. Eligibility involves full-text evaluation based on the criteria. Inclusion includes the final studies that relate to the topic, with filtering organized to retain only high-quality studies. The PRISMA flow in Figure 2 shows how the studies were identified, screened, assessed and selected for this review. In the first stage, a total of 520 records were found from Scopus, IEEE and Web of Science. 460 studies were left after the removal of 60 duplicate records. During the title and abstract screening 227 publication were excluded because they address ineligible topics and were not published with English. Leaving 233 articles which were further subject to full-text assessment. At the eligibility stage 153 publications were excluded because they did not relate to the human factor or were theoretical studies that did not have an empirical application. This resulted in 80 studies being included in the systematic review.

The quality assessment criteria used to evaluate each of the studies include methodological rigor, data reliability, AI model transparency, and focus on human/organizational factors. This is necessary since PdM research is usually challenged by the quality of data and complexity of systems. Also, the integration of the industrial psychological perspective is one of the unique contributions of this SLR, with an emphasis on human trust in AI systems, cognitive workload in decision-making, organizational readiness and training, and human–AI collaboration in maintenance environments. This is in line with the new studies that place human factors in predictive systems powered by AI. Therefore, the ethical considerations are citation of all sources, no bias in study selection and disclosure of limitations.

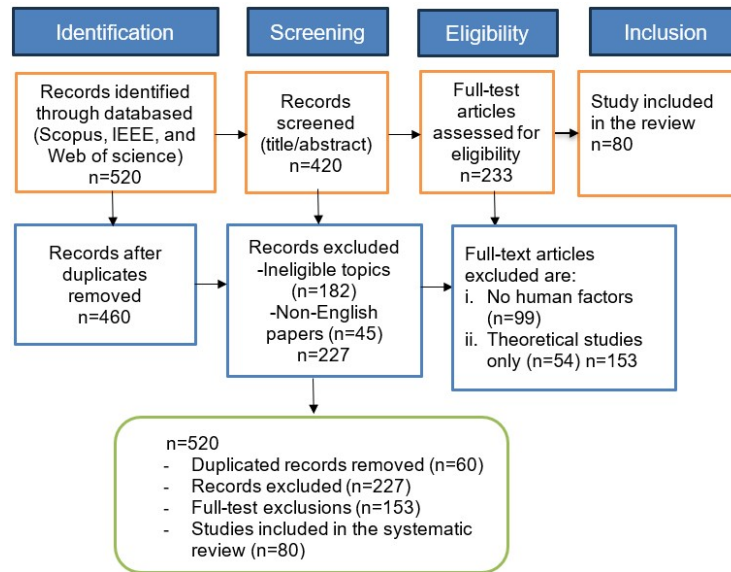


Figure 2. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses flow diagram optimising renewable energy infrastructure through AI-driven predictive maintenance and fault detection

3 Results Analysis From the Literature of Related Study

3.1 Evolution of AI in Renewable Energy

Only renewable energy sources, such as biomass (in the form of wood) for cooking, heating, and lighting, wind energy for navigation and mill power, and hydropower for mill power, were available before the introduction of fossil fuels. According to Qin et al. [18], investigating the vital role of AI in the energy industry is key to advancing the growth of renewable energy. To determine the relationship between China's artificial intelligence index (AII) and the renewable energy indicator (REI), the study employs both full and subsample approaches. It is evident from the quantitative debates that AII has both good and negative effects on REI, with the former suggesting that AII encourages the development of renewable energy. However, if AII has a negative impact on REI—primarily due to less expensive non-renewable energy—this incentive role cannot be established in the long run. Conversely, during the COVID-19 pandemic, REI had a positive impact on AII, demonstrating how the decline in financial markets and renewable energy brought on by the pandemic hindered China's advancement in AI.

Lyu et al. [19] explored the development of renewable energy that has attracted the interest of numerous academics, stakeholders, and government over the past years. Nevertheless, the possible role of AI in advancing renewable energy has not been identified by the previous literature. To start with, the main conclusion, that AI is an important factor in the enhanced development of renewable energy, is backed up by two robust tests, where the independent and dependent variables are varied. Second, the connection of AI with the creation of renewable energy is closer in the countries where the development of renewable energy is less. AI, as a process in the development of renewable energy, deals with innovation and technological development. Fourth, climate funding directly contributes to the growth of renewable energy, and it also serves to regulate the relationship between the evolution of renewable energy and AI. The integration of AI has become one of the primary forces towards the enhancement of production, minimization of costs, and the resolution of difficult issues in the innovative and progressive field of sustainable and renewable energy. Nonetheless, none of the sources can cover all the developments covered in recent years, their experimental applications, and the barriers that accompany them [20]. As Fan et al. [21] highlighted, AI has already been found to be a helpful resource in the renewable energy sector to enhance the detection of problems, management of energy, and stability of the power grid. Also, they have demonstrated potential in improving prediction analysis of waste management and photovoltaic power plants. In environmental health it can be used to improve exposure modelling and disease prediction and simplify the analysis of complex spatial data. However, the difficulties with the explainability and transparency of AI models, scalability and high dimensionality of data, compatibility with the next-generation wireless networks, and ethics and privacy should be resolved. Future research should be aimed at making AI models explainable and transparent, creating scalable paradigms to support large data volumes, exploring the collaboration between AI and next-generation wireless networks, and ethical and privacy issues. The AI models also need to be made more energy-efficient and the sustainability of such systems should be enforced long-term.

Hamdan et al. [22] provided a critical examination of large amounts of sensor and prior performance information being processed by AI to define the trends that can signify problems. Other than reducing the times of non-

availability, such a proactive idea increases the lifespan of the renewable power infrastructure, which factor makes the cost significantly lower and more dependable. Along with this, AI should be utilized to streamline the utilization of renewable sources of energy. The changing environment can be adjusted to with the help of AI systems, which can screen in real-time and analyze data with the help of advanced analytics to predict the patterns of energy production and resource allocation with minimal resources utilization. By doing so, renewable sources of energy can guarantee that they produce the maximum amount of energy and hence can compete with the conventional sources of energy. The article authored by Kumar and Swathika [23] provides a review of AI applications in the renewable energy sector, explaining how electrical engineering has been revolutionized using AI. Their analysis shows how AI and RETs affect the generation, transmission, and use of energy. The subsection tells what AI may add to designing smart utilities and buildings, and how AI, together with renewable energy, may cooperate to create smarter and more efficient energy systems. These models can be constantly improved by training on previous data from different projects on renewable energy and can adapt to new input data. The most significant findings suggest that Lifecycle assessment (LCA) models driven by AI can enhance the accuracy and comprehensiveness of environmental impact evaluations.

According to Manuel et al. [24], AI technologies and ML and predictive analytics are capable of greatly improving distribution and overall efficiency of a RES due to their role in contributing to optimal energy production, enhancing system performance and fostering faster and safer processes. AI PdM utilizes renewable energy assets, such as wind turbines and solar panels, to predict and preempt equipment issues by analyzing vast quantities of data generated by them. This proactive strategy will guarantee the efficient and consistent production of energy because of the minimization of maintenance expenditures and unavailability. An AI system also predicts the energy output and demand, which optimizes distribution and storage of energy correctly. This optimisation would decrease the use of non-renewable energy supply because it would make the grid more stable and balance the supply and demand of electricity products better. The past couple of years have seen an increasing intensity and scope of deployment of AI and ML solutions in the RES.

Identifying and applying the most efficient approaches to the continuation of the study may be challenging because ML and AI have numerous applications. To determine the solution to this problem, this study analyzes some of the most popular and renowned AI solutions to renewable energy. The description of particle swarm optimization (PSO), recurrent neural networks (RNNs), and convolutional neural networks (CNNs), artificial neural networks (ANN), Long Short-Term Memory (LSTM), genetic algorithm (GA) and over ten most popular through the day RES modelling and optimization algorithms can be found in the given paper [25]. Figure 3 summarizes the different functions that AI would be able to fulfill in improving the effectiveness of Renewable energy [26].

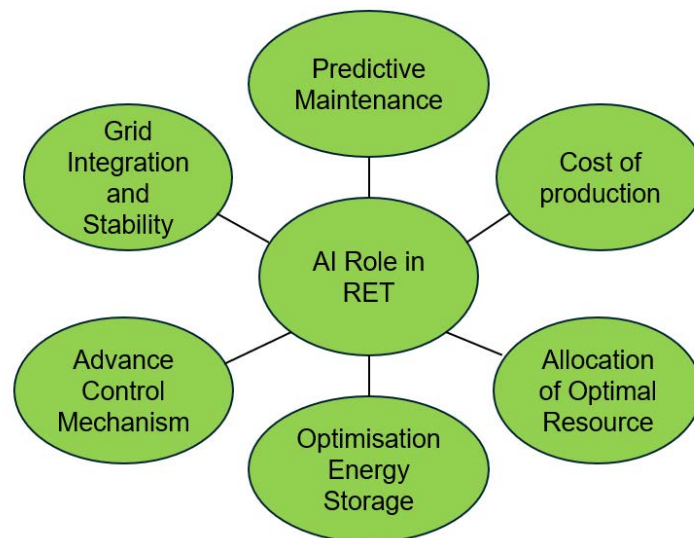


Figure 3. Schematic design of the role of AI in enhancing the proficiency of renewable energy, based on Ref. [26]

Figure 3 shows the main application and advantage of AI in RET. The core is the central role of AI in improving, managing, and optimizing today's green energy systems in 6 major fields:

- (i) PdM: AI models process real-time operational data to predict equipment wear and potential failures before they happen, reducing downtime and repair cost.
- (ii) Cost Reduction: Automated monitoring and smart optimization lower the energy operational cost and inefficiencies throughout energy assets.

- (iii) Optimal Resource Allocation: ML models estimate power generation (such as solar irradiation or wind speed) and match power generation-in-demand to allocate resources optimally.
- (iv) Energy Storage Optimization: smart control systems optimize battery and storage capacity charging and discharging to maximize utilization and lifespan.
- (v) Advance Control System: AI for real-time dynamic control of complex power generation equipment to always optimize performance.
- vi. Grid Integration and Stability: Intelligent Forecasting provides control of the intermittent renewable inputs to ensure grid stability and prevent frequency and voltage variations [26].

3.2 AI-Driven Fault Detection in Renewable Energy Systems

Pradeep et al. [27] states that to enhance operating efficiency and grid resilience, new methodologies for problem detection and PdM are necessary, as electrical power systems are becoming increasingly complex. Traditional maintenance methods, such as reactive maintenance and preventive maintenance, have proven insufficient to reduce unplanned outages and optimize the utilization of assets. Recent advances in AI involve data-driven solutions that significantly enhance automated recovery processes, failure prediction, and fault classification [28]. The advantages of using AI-based defect finding over traditional methods are demonstrated through case studies and simulations. Such advantages include a major improvement in terms of functional eliciting ability, delicacy, and reaction time. Investigated why, because of high-dimensional data, measuring noise, and slow fault diagnosis processes, the expansion of modern power systems, RES, and decentralized power systems is problematic to the traditional fault detection methods because of their difficulty in fault detection techniques [29]. It suggests the deep learning of failure identification in the smart grid using image-detected Phasor Measurement Units (PMUs). The proposed approach facilitates effective fault classification by transmitting PMU data into images and utilizing complex deep learning models, including VGG16 and CNNs. In fault classification, the experimental findings show that VGG-16 had an accuracy of 98.75%, whereas CNNs had an accuracy of 94.44%. According to Somanna et al. [30] among the flaws that the model will identify within power system faults without error selectivity, there are short circuits, line-to-line faults, and three-phase faults, as well as others. Such errors may lead to equipment breakdown, power failure, and safety risks. In simulation, the system effectively utilizes significant parameters, such as voltage drops, current surges, and power quality oscillations, by simulating the fault state in Simulink. The model is parameterized to simulate realistic power systems by including typical components of a power system, such as transformers, circuit breakers, fault blocks, and a three-phase source. Fault response and proper fault detection become feasible using monitoring techniques that measure system behavior during faults. The proposed system features an effective problem identification process, timely protection strategies, and the most suitable recovery system methods.

According to Joshua et al. [31], the main industries that fully depend on renewable energy must have adequate supply of solar power that would maintain their operations effectively. However, in cases where weather conditions, equipment wear and tears and inefficiencies in the system cause changes in the quantity of solar energy produced, a consistent supply of power would be difficult. The fluctuating energy production has rendered an alloy manufacturing firm to achieve its expectations in production which explains why an elaborate anomaly detection and performance optimization system is required. The deficiencies of the solar infrastructure are unknown and might cause waste of energy, reduced productivity, and disruption of operations, which may adversely affect the production of the industry at a greater scale. To improve the reliability and strength of solar electricity, he suggests a framework-based anomaly detection that is powered by AI. The study, conducted by Hosseinzadeh et al. [32], provides an AI-powered model that detects faults in solar panels on the solar-hydrogen system implemented at the Samcheok Campus of Kangwon National University.

Ensuring the optimal performance and reliability of RES in universities and other educational institutions is becoming increasingly important. They suggest using automatic deep learning architecture, specifically a ResNet-50 model, to identify defects in solar panels. The data from the solar-hydrogen system at the campus was pre-processed to optimize the ResNet-50 model for correctly classifying faults. The works by Bello et al. [33] the paper enables educational institutions to adopt sustainable energy management and lays the groundwork for future studies on implementing AI-powered defect diagnostic mechanisms in RES. As indicated in Figure 4, the understudy smart grid comprises a 100-kW solar power plant, a 5-kW energy storage system, a 20-kW fuel cell system, a linear load of 800 kW, ± 200 kvar, and a nonlinear load and power system. The nonlinear load power ranges from 100 to 500 kW, and the power factor varies. With this smart grid, it is possible to represent all the required faults and provide the necessary data to develop a comprehensive fault detection system [34].

3.3 Predictive Maintenance Using AI for Optimized Performance

Yousef et al. [35] address the current state of AI in the renewable energy sector and its impact on energy optimization and PdM across various energy sources, including solar, wind, and hydro. By examining the common AI techniques employed in the field under consideration, they describe how AI can help ensure that RES remain sustainable and efficient. The fields of infrastructure maintenance, energy generation optimization, and renewable

source integration within the grid are also being revolutionized by AI [36]. It's optimized, predictive, and advanced analytics that will help the world reach its renewable energy goals. The sustainable future will become increasingly cleaner as AI technology continues to disrupt the renewable energy industry. Because of what AI makes possible, we can move faster toward a clean energy world and leave behind a habitable planet for our children to inherit.

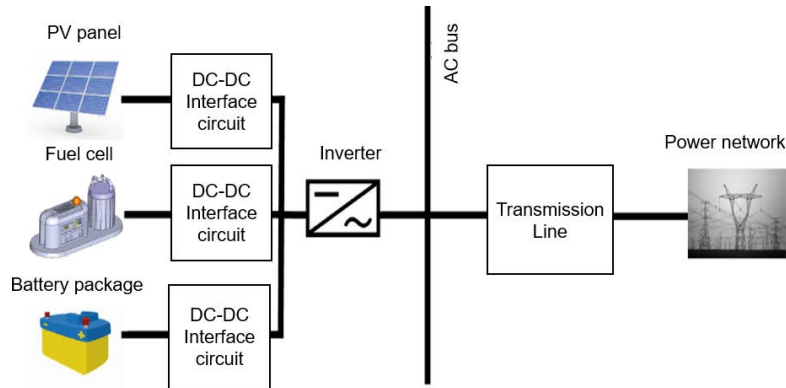


Figure 4. Detection and categorization of faults in smart grids by an augmented K-NN algorithm
 Note: DC = direct current; PV = photovoltaic; K-NN = k-nearest neighbors.

AI is needed to optimize the energy produced by renewable energy sources. Due to advanced data analytics and real-time results observation, AI mechanisms can dynamically adjust to changes in environmental conditions while predicting the trends in energy production and allocating resources as efficiently as possible [37]. The AI-PdM comprises six major components, which are as follows: data pre-treatment, AI technique, decision-making modules, integration and communication modules, user interface, and reports. These can be explained as shown in Figure 5.

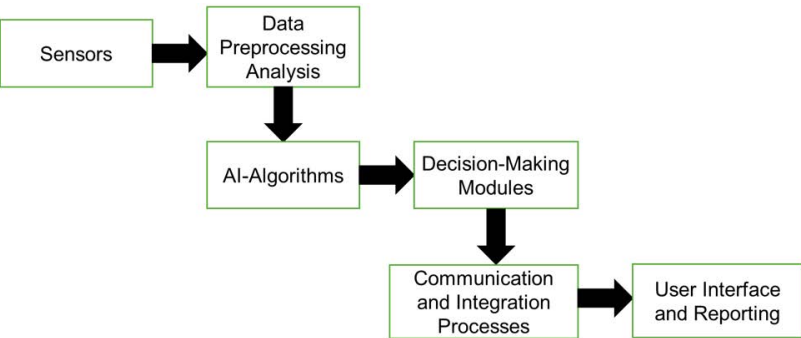


Figure 5. Key components of an AI-based predictive maintenance system, adapted from [37]

3.4 Descriptive Analysis of the Systematical Review Study

Table 2 shows the summary of the most recent findings in the field of PdM and AI-based fault detection across various industries. It describes various models and methods, including ML, deep learning, LSTM, GAs, digital twins, and Industrial Internet of Things platforms, which are applicable in diverse Industry 4.0 applications, power networks, maritime systems, facilities management, and industrial automation. Overall, the articles have revealed that PdM implemented by AI can be used to augment the quality of fault detection, reduce downtime, streamline maintenance processes, and enhance productivity and cost efficiency [38]. Table 2 also shows that there is a growing interest in new trends, such as edge computing, digital transformation, and human-technology integration, which provide strategic importance of AI in contemporary maintenance systems.

The comparison in Table 3 shows that there is no one size that fits all solutions for renewable energy PdM using AI. Structured datasets like classification or maintenance benefit from conventional ML methods like SVM and Random Forest, while deep learning methods like CNN and LSTM work well with complex spatial and temporal patterns. In unsupervised anomaly detection, when data on failures are unavailable, autoencoders offer opportunities, and in adaptive scheduling, reinforcement learning can assist with scheduling decisions.

However, the hybridization of AI and digital-twin approaches brings with it increased levels of integration and optimization, while also creating further computational, interoperability, explainability and implementation challenges. Importantly, these methods must be carefully selected and successfully implemented, considering

predictive accuracy, but also interpretability, employee trust, employee competency, cognitive workload and organizational readiness. This humanistic approach is very important in the industrial psychological dimension of the present study.

Table 2. Summary analysis of PdM using AI for optimized performance

Author(s)	Model/Method	Application Area	Key Findings
Keleko et al. [39]	The bibliometric analysis tools used are VOSviewer and Power BI	Industry 4.0 (I4.0)	AI-driven PdM4.0 contributes to improved productivity through enhanced availability, quality, cost reduction, early fault detection, and reduced downtime.
Gadde [40]	AI techniques applied: ML algorithms; Data analytics	Relational database systems	Introduces a novel application of AI in PdM for relational databases.
Bidollahkhani and Kunkel [41]	Survey/review paper ML; neural networks	Computing continuum systems (encompassing cloud, edge, and fog computing)	The survey identifies advancements, methodologies, and challenges in AI-driven PdM. AI, especially ML and neural networks, improves failure prediction accuracy and maintenance scheduling in complex distributed environments.
Kim and Choi [42]	LSTM; GA	PdM for machinery or industrial equipment, focusing on: prognostics; health monitoring; model-driven decision-making	The LSTM-based PdM model is enhanced by reducing irrelevant features and optimizing model structure. The GA effectively tunes LSTM hyperparameters, improving the model's prediction accuracy and efficiency.
Simion et al. [43]	AI and ML	Maritime industry/naval systems	Effectiveness of AI/ML in Maritime Maintenance: AI and ML techniques can accurately diagnose faults, predict failures, and optimize maintenance schedules.
Babu et al. [44]	Advanced ML algorithms. Edge computing integration. Digital twin technology	Electrical equipment and power networks	Edge computing and digital twins enable real-time diagnostics and forecasting. Emphasis on emerging trends, including digital transformation, AI ethics, and tighter integration with Industry 4.0 strategies.
Raj et al. [45]	A comprehensive range of AI, ML, and deep learning models, including CNN, RNN, LSTM, fuzzy logic, and SVM, as well as blockchain and edge AI technologies.	IT energy efficiency, green AI, environmental sustainability, smart Grids, healthcare, fashion, and digital lean manufacturing	AI is a transformative force that significantly enhances energy demand forecasting, optimizes resource allocation, and improves fault detection (e.g., Identifying photovoltaic panel faults with >98% accuracy). The integration of edge AI and IoT facilitates real-time, remote diagnostics and reduces operational and maintenance costs.
Koumoulos et al. [46]	IoT-based frameworks utilizing microcontroller platforms (ESP32, NodeMCU) and single-board computers (Raspberry Pi) integrated with optimization algorithms (PSO, WOA-SA) and neural models (ANN, LSTM, CNN-LSTM).	Off-grid solar systems and autonomous microgrids, including agrivoltaics applications	IoT-based architecture significantly improves energy efficiency and enables PdM. Hybrid edge-cloud intelligence effectively balances computational complexity with power constraints. Digital Twin implementations achieved a 22% yield gain in agrivoltaic systems.
Sarker [47]	A hybrid framework integrating quantum computing algorithms (e.g., QAOA) with deep reinforcement learning (e.g., DDPG, PPO) and IoT-enabled data acquisition.	Real-time optimization and power quality management in renewable energy-driven smart grids	Synergizing quantum parallel processing with AI diagnostics enables rapid solutions to complex grid problems. The framework reduced voltage sag by up to 39%, enhanced active power transfer by approximately 38.77%, and improved overall grid resilience and cybersecurity through quantum-safe encryption.

Note: PdM = predictive maintenance; ML = machine learning; LSTM = Long Short-Term Memory; GA = genetic algorithm; PSO = particle swarm optimization; WOA-SA = Whale Optimization Algorithm Simulated Annealing; CNN = convolutional neural network; RNN = recurrent neural network; SVM = support vector machine; ANN = artificial neural network; DDPG = Deep Deterministic Policy Gradient; PPO = Proximal Policy Optimization.

Table 3. Comparison of AI methods for predictive maintenance and fault detection in renewable energy infrastructure

AI Method	Typical Applications	Key Strengths	Major Limitations
Machine learning	Fault classification, condition monitoring, failure prediction	Effective for structured datasets; relatively simple implementation	Performance depends strongly on feature engineering and data quality
Support vector machine	Fault classification and anomaly detection	Effective with small and high-dimensional datasets	Computationally demanding for large datasets; sensitive to parameter selection
Random forest	Fault diagnosis, component failure prediction	Robust, interpretable, and resistant to overfitting	Less effective for highly complex temporal patterns
Artificial neural networks	Wind and solar power prediction, fault detection	Captures nonlinear relationships	Requires substantial training data; limited interpretability
Deep learning	Complex fault diagnosis, image-based inspection, anomaly detection	Automatically learns complex features and nonlinear patterns	High computational requirements and limited explainability
Convolutional neural networks	Solar panel image inspection, visual fault detection	Highly effective for image and spatial-pattern recognition	Requires large, labelled datasets and computational resources
Long short-term memory	Time-series prediction, degradation and failure prediction	Effective in modelling temporal dependencies	Training can be computationally intensive; requires substantial historical data
Autoencoders	Unsupervised anomaly and fault detection	Useful when labelled fault data are limited	Reconstruction-based results may be difficult to interpret
Reinforcement learning	Maintenance scheduling and operational optimization	Enables adaptive decision-making and optimization	Requires carefully designed reward functions and extensive training
Hybrid AI models	Integrated prediction, fault diagnosis and maintenance optimization	Combines complementary strengths of multiple algorithms	Greater complexity, computational requirements, and validation challenges
Digital twin + AI	Real-time monitoring, predictive maintenance and asset-life prediction	Enables continuous virtual-physical system interaction	High implementation cost, data integration challenges, and interoperability issues

4 Discussion of the Industrial Psychological Perspective

The research areas of Industrial and Organisation (I/O) Psychology offer a holistic analytical standard of how the AI PdM, and the fault detection can affect the behavioral aspects of humans, organizational process, and overall process performance in the context of renewable energy infrastructure [48]. Even though AI technology enhances technical reliability due to the ability to detect in the early stages the faults that occur and make decisions related to information based on maintenance, to the level of the implementation, is highly dependent on the interaction human-technology, workers attitude, and preparedness of the organization [49]. I/O Psychology shows that both theoretical and empirical analysis are considered for non-technical areas, so that AI-based systems of maintenance could be not only competent in terms of technology, but also psychologically likeminded and operational efficient [50]. Therefore, at a personal level, I/O psychology deals with the problem of the operators trust in the recommended AI, cognitive workload, awareness that are situational, the uncertainty under the decision-making. Even the maintenance predictive model may demand the operators or specialists comprehend the generative AI warning and projections, which makes the human parameters a key element of the reliability system [51].

Over-automation or veiled algorithms can be transformed into overreliance, non-utilization or resistance, and thus diminish the efficiency of the system. I/O psychology assists in evaluating such risks and informs human-centred AI interfaces, which foster instead of displacing human skills. The organizational level of I/O psychology analyzes the impact of leadership support, training practices, safety culture, and change management practices on the implementation of AI-based maintenance strategies [52]. Reactive to PdM transition is an important organizational change that changes employment functions, competencies, and responsibility frameworks. Using the principles of

I/O psychology, organizations will be able to more effectively deal with the adaptation of their employees, improve their skills, and learn to accept AI technologies. Overall, the placement of I/O psychology as the analytical prism allows viewing of the PdM and AI-based fault detection in their entirety by considering the elements of technical performance, human, and organizational factors to optimize the renewable energy infrastructure sustainably [53].

Figure 6 illustrates a conceptual framework linking AI systems, human factors, and organizational outcomes in the context of fault detection and PdM for renewable energy infrastructure.

(i) AI Systems (Top Layer): This layer encompasses technological enablers such as AI-powered fault detection, AI-enabled PdM, and AI-enabled real-time monitoring & analytics. These systems function as inputs, providing data and alerts, as well as PdM insights to aid in maintenance operations [54].

(ii) Human Factors (Middle Layer): This layer considers the human component in the interface with AI systems, and addresses three concerns:

- Human–AI Interaction & Trust: The degree of trust, and belief of the operators in AI, and the AI recommendation [55].
- Cognitive Workload & Decision-Making: The degree to which AI influences cognitive load, situation awareness, and the quality of operational decisions.
- Employee Attitudes & Skills: The degree of workforce preparedness, the level of employee capability, and the degree of employee willingness to work with AI.
- This layer serves as an intermediary, demonstrating that the operationalization of AI systems is dependent on human input and engagement [56].

(iii) Organizational Outcomes (Bottom Layer): This layer captures the effect of AI-human integration on the organizational performance, which includes

- Operational Efficiency: The extent of maintenance downtime and schedule optimization.
- Safety & Risk Reduction: The extent to which accidents or equipment failures are avoided.
- Employee Well-Being & Engagement: The extent to which job satisfaction increases and work-related stress decreases [57].
- Organizational Change & Culture: The extent of leadership sponsorship, AI practice adoption, and the change in culture towards innovations.

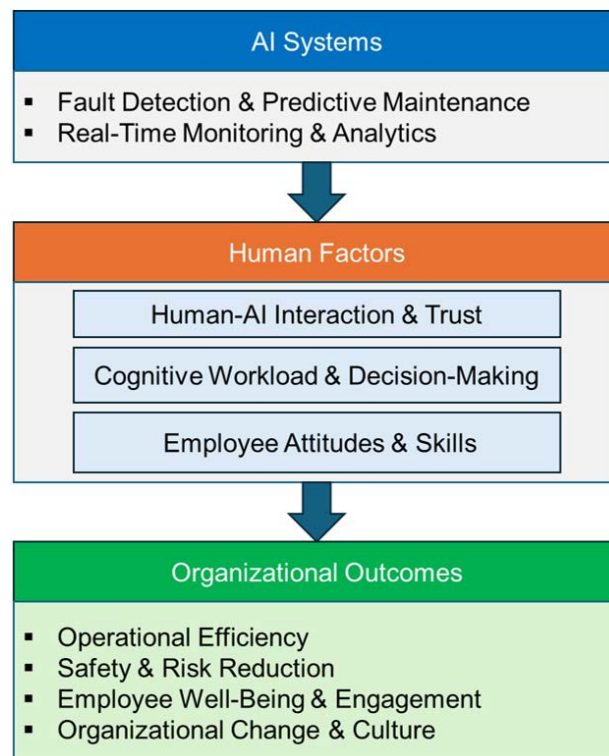


Figure 6. Conceptual framework linking AI-driven fault detection and predictive maintenance, human factors, and organizational outcomes

Figure 6 demonstrates a top-down approach where “At the top, AI systems give the most insight, and at the bottom, human factors help the most ensure effective use and trust, leading to organizational success in optimizing applications in renewable energy infrastructure.” This approach highlights the importance of human-technology collaboration in

optimizing AI-powered maintenance systems to gain maximum benefits from the maintenance systems in optimizing renewable energy infrastructure [58]. In relation to fault detection and PdM, Table 4 offers a comprehensive synthesis of the existing literature concerning the AI-driven systems, human factors, and organizational outcomes triad focused on optimizing renewable energy infrastructure. Table 4 is structured to guide the reader to the most relevant literature and applicable insights regarding the use of AI technology in an industrial context.

Table 4. Industrial psychological dimensions of AI-driven fault detection and PdM in optimizing renewable energy infrastructure

Theme	Author(s) & Year	Study Focus/Findings	Key Insights	Relevance to AI-Driven Fault Detection & PdM
Human-AI interaction and trust	Doh et al. [59]	Examined trust calibration in automated systems; overreliance and under-reliance can reduce performance. Studied AI decision support in manufacturing; trust influenced by system transparency and reliability.	Trust management strategies are critical to optimize human-AI collaboration.	Ensures operators trust AI fault alerts and predictive recommendations, avoiding over/under-reaction.
	Ahangar et al. [60]	Examined trust calibration in automated systems; overreliance and under-reliance can reduce performance. Studied AI decision support in manufacturing; trust influenced by system transparency and reliability.	Transparent AI explanations improve user confidence and operational adherence.	Supports clarity of AI-generated maintenance alerts for accurate human decision-making.
	Amaliah et al. [61]	Explored the effect of AI errors on operator trust in PdM systems.	Error communication and feedback loops maintain operator trust.	Reduces downtime caused by mistrust in AI predictions and improves adoption of AI maintenance tools.
Cognitive workload and decision-making	Lyu et al. [19]	Reviewed workload in automated systems; high automation can reduce situational awareness.	Balancing automation with human monitoring enhances decision-making.	Helps design PdM dashboards that prevent operators from overloading and errors.
	Hauptman et al. [62]	Examined adaptive automation and operator cognitive load.	Adaptive AI interfaces reduce cognitive strain while supporting complex tasks.	Optimizes human oversight of fault detection processes, improving response accuracy.
	Morić et al. [63]	Investigated AI support in energy management; measured decision accuracy under workload conditions.	Well-designed AI dashboards reduce cognitive load and improve accuracy.	Enhance operators' ability to interpret AI-driven PdM alerts efficiently.
Employee attitudes, skills, and acceptance	Venkatesh et al. [64]	The Technology Acceptance Model applied to AI tools in industrial settings.	Training and perceived usefulness are key predictors of AI adoption.	Guides workforce training programs to improve the adoption of AI fault detection systems.
	Abrar et al. [65]	Studied industrial workforce readiness for AI-driven maintenance.	Skills development enhances confidence and acceptance.	Ensures staff competence in using PdM tools for proactive interventions.
	Li et al. [66]	Investigated user attitudes toward AI PdM in energy systems.	Positive attitudes correlate with greater reliance and engagement with AI systems.	Directly relevant to ensuring effective human-AI collaboration in maintenance decisions.

Note: PdM = predictive maintenance.

This study discusses that PdM and fault detection driven by AI can support the improvement of efficiency of the operational processes of renewable energy infrastructures (solar, wind, and hybrid systems) [67, 68]. Acceptance, performance, and trust in these advanced technologies were affected by the industrial psychological elements like employee attitude, cognitive readiness, and behavioral adaptation [69, 70]. Companies that combine human elements with the technical side of things also tend to do better about system uptime, safety and cost-effectiveness [71]. Moreover, the use of AI in maintenance processes enables the operational staff to act ahead by driving down unplanned downtimes and improving the attitude towards continuous improvement. Furthermore, the study uses wind energy and solar energy illustration to express real life practical engineering knowledge

AI for wind turbine PdM: AI-driven PdM solutions can continuously monitor wind farm operations to detect early signs of wear and tear on wind turbines, such as vibrations, temperatures, gearbox wear, generator issues, and other condition metrics [72]. But the usefulness of these systems is contingent on the maintenance engineers' capacity and confidence to believe and act on the AI warnings. Proper training, clear AI-generated outputs, and seamless human-AI teamwork can boost engineers' trust in AI suggestions and facilitate timely maintenance decisions. On the other hand, if trust is low, technical skills are not up to scratch or resistance to technological change, the AI recommendations may be overlooked, limiting the benefits of PdM [73].

Solar photovoltaic fault detection: For large scale solar photovoltaic systems, AI and computer vision systems can use sensors and imaging data to identify anomalies like abnormal operating conditions, and faulty modules. However, the alerts need to be interpreted by the maintenance personnel, the problem confirmed, and the correct maintenance action taken [74]. When employees possess the necessary AI skills and have access to organizational resources, an AI-generated alert can help identify faults and plan maintenance in a timely manner. But if the use of automated recommendations is excessive, employees might not be sufficiently competent or might have a high cognitive load, which could impact on the quality of the decision [75]. This highlights the significance of bringing together the capabilities of AI with human skills and organization to ensure the best results. The practical examples in this review reflect the main objectives of this study that for optimizing renewable energy infrastructure, AI solutions are needed that are both technically correct and capable of engaging with humans through human-AI interaction, preparing personnel for the shift to AI-based maintenance tools, and creating organizational conditions for the uptake, and continued utilization, of AI-based maintenance technologies.

5 Future Research Directions

The results of this SLR suggest that future research ought to shift away from technology-focused research and towards a more socio-technical approach. Based on this, future research opportunities are grouped into the technical, organizational, and human/industrial psychological dimensions.

5.1 Technical and Technological Research

To enhance the technical aspects of AI-based PdM and fault detection systems for renewable energy infrastructure, future studies should be conducted [76]. Research is required to create more interpretable, capable, flexible, and generalizable AI models that can work across varying environmental and operational conditions. Hybrid AI models, deep learning, digital twins, edge computing, and real-time fault diagnosis are all important considerations. The use of multi-source sensor data, IoT systems, supervisory control and data acquisition, weather information, and historical maintenance records should also be investigated for future research to enhance prediction accuracy. AI systems should be benchmarked against well-known datasets and metrics for comparative analysis [77]. Further, studies must investigate the following aspects of data quality, cyber-security, model transferability, interoperability, and explainability of AI-generated maintenance recommendations.

5.2 Organizational and Management Research

Future studies should focus on the challenges of integrating AI-based PdM into existing maintenance and asset management systems, and how these systems can be made effective. Research into organizational readiness, leadership support, resources, digital transformation strategies, culture innovation, training policies, and change-management practices should be engaged [78]. Longitudinal and multi-site studies would clarify the adaptation of organizations over time in the use of AI in maintenance. Future studies should also explore the economic impact of AI adoption, such as adoption costs, return on investment, maintenance savings, improved productivity, and benefits for the life cycle of AI systems [79]. An examination of case studies of RES in different geographical areas and renewable energy sectors might reveal more contextual factors that impact the implementation.

5.3 Human and Industrial Psychology

One of the key research agendas is to explore the human factor in AI-powered PdM. Future research should investigate employee trust in AI, technology acceptance, human-AI interaction, cognitive workload, job satisfaction, perceived autonomy, job competence, resistance to technological change, and safety behavior [80]. This is

especially important for empirical research to include maintenance engineers, maintenance technicians, operators, and managers, as they are key players in the field, with much of the literature being technology-driven. Surveys, interviews, experiments, and longitudinal designs can be employed in future studies to understand employees' responses to AI-generated maintenance recommendations and the impact on operational performance and decision-making.

5.4 Integrated Socio-Technical Research

Finally, future studies should explore the relationship between technical, organizational, and human factors. Future studies should build on AI performance, organizational readiness, and employee attitudes to develop a comprehensive model that connects these aspects together to predict how they influence the optimization of renewable energy infrastructure [81]. This research could help shape the evolution of human-centered AI PdM systems that boost technical reliability, organizational performance, employee productivity, and sustainable energy infrastructure results.

6 Conclusions and Recommendations

This paper examined the potential of optimizing the renewable energy infrastructure through AI-based fault detection and PdM. It also integrates the framework of industrial psychology, which studies the interaction of people and technology in the workplace. The findings indicate that AI can enhance the efficiency, reliability, and sustainability of RES, and that human elements are critical for the adoption and implementation.

(i) Enhanced Accuracy of Fault Detection: AI algorithms, particularly ML and deep learning, were remarkably effective in predicting both anomalies and failures, resulting in timely interventions and reduced operational disruption.

(ii) Improved Efficiency of PdM: The implementation of PDM resulted in reduced unplanned maintenance, as equipment failures were anticipated, resources were effectively utilized, and the operational life of critical assets was extended.

(iii) Industrial Psychology: Employees' attitudes towards AI adoption impacted the system's utilization. In the presence of favorable perceptions of AI, which were enhanced through training and participatory decision-making, compliance and operational performance improved significantly. Behavioral Adaptation: The researchers noted that employees who were trained in AI tools in a structured manner showed greater engagement, less resistance to change, and enhanced cooperation with automated systems.

(iv) Organizational Benefits: The Organizations that employed AI-based maintenance indicated cost-cutting, enhanced reliability, and safer work, underlining that the technical method proves to be the most effective in the case of coordination with the behavioral elements of humans.

Therefore, to implement it successfully, it is necessary to invest in both technology and human resources, focusing on employee training, ergonomic systems, and continuous feedback systems to maintain motivation and trust. The technical and psychological synergy of investigating renewable energy infrastructure using PdM and AI-based fault detection optimizes the process. With technological efficiency and human factors in mind. Organizations can optimize system performance, make their operations sustainable, and develop an adaptive workforce capable of effectively utilizing AI innovations.

Author Contributions

Conceptualization, I.P.O.; methodology, I.P.O.; software, I.P.O. and L.T.; validation, I.P.O. and L.T.; formal analysis, I.P.O.; investigation, I.P.O.; resources, I.P.O. and L.T.; data curation, I.P.O. and L.T.; writing—original draft preparation, I.P.O.; writing—review and editing, L.T.; visualization, I.P.O.; supervision, L.T.; project administration, L.T.; funding acquisition, L.T. All authors have read and agreed to the published version of the manuscript.

Data Availability

Not applicable.

Conflicts of Interest

The authors declare no conflicts of interest.

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